

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – CASE STUDY

|                  |                                                                                                          |               |                                   |
|------------------|----------------------------------------------------------------------------------------------------------|---------------|-----------------------------------|
| Course Code:     | CSA06                                                                                                    | Course Title: | Design and Analysis of Algorithms |
| CO Assessed:     | CO3: Articulation (BL4, BL5)                                                                             | Assessment:   | Assessment Tool 2 – Case Study    |
| Weightage:       | 20% of CIA                                                                                               | Total Marks:  | 100 (2 Questions × 50 Marks)      |
| CO3 Statement:   | Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication |               |                                   |
| Student Name:    | KS THARUNESHWAR                                                                                          | Reg. No.:     | 192572148                         |
| Section / Batch: | 2025                                                                                                     | Date:         | 18-08-2026                        |

### Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues, in-depth analysis, Innovative, feasible solutions with strong justification

| Question Type | Description                                                                                                    | Questions | Total Marks |
|---------------|----------------------------------------------------------------------------------------------------------------|-----------|-------------|
| Case Study    | Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication | Q1 – Q2   | 100         |



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DATA PROGRAMMING



V Deepa Dharshini

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CO3-AT3-Case Study

Language:

Python

C

C++

Java

QUESTIONS

Q1

Divide and Conquer

50 marks

Q2

Divide and Conquer

50 marks

2/2 answered

Q1 Divide and Conquer

50 marks

Case Study: Engineering Simulation System (Matrix Multiplication)

An engineering simulation tool performs repeated matrix operations for structural analysis. Analyze how divide and conquer matrix multiplication improves computational efficiency. Identify challenges in large-scale matrix handling. Apply optimized methods such as Strassen's algorithm. Analyze performance and design a solution, justifying its advantages.

Your Code

```

1 def add_matrix(A, B):
2     n = len(A)
3     return [[A[i][j] + B[i][j] for j in range(n)] for i in range(n)]
4
5
6 def subtract_matrix(A, B):
7     n = len(A)
8     return [[A[i][j] - B[i][j] for j in range(n)] for i in range(n)]
9
10
11 def strassen(A, B):
12     n = len(A)
13
14     if n == 1:

```

Input

```

2
1 2
3 4
5 6

```

Output

```

Enter matrix size: Enter
Matrix A:
Enter Matrix B:

Resultant Matrix:
43 22

```

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DAA PROGRAMMING

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## CO3-AT3-Case Study

← Back

Language: Python C C++ Java

### QUESTIONS

Q1

Divide and Conquer  
50 marks

Q2

Divide and Conquer  
50 marks

2/2 answered

### Q2 Divide and Conquer

50 marks

Case Study: Banking Transaction Processing (Merge Sort)

A banking system processes and sorts millions of daily transactions based on timestamps for auditing and reporting. Analyze how Merge Sort can be applied to ensure stable and efficient sorting. Identify key issues such as large data volume and need for stability. Apply divide and conquer concepts to explain the approach. Analyze time complexity and performance. Design a suitable solution and justify its effectiveness in this scenario.

#### Your Code

```
1 def merge_sort(transactions):
2
3     if len(transactions) <= 1:
4         return transactions
5     mid = len(transactions) // 2
6     left = transactions[:mid]
7     right = transactions[mid:]
8     left = merge_sort(left)
9     right = merge_sort(right)
10    return merge(left, right)
11
12 def merge(left, right):
13     result = []
14     i = 0
```

#### Input

```
5
26-08-18-14:30 T103 5000
2026-08-18-09:15 T101 2500
2026-08-18-12:45 T102 3000
```

#### Output

```
Transactions sorted by
timestamp:
2026-08-18-09:15 T101 2500.0
2026-08-18-11:10 T104 1500.0
2026-08-18-12:45 T102 3000.0
2026-08-18-16:20 T105 7000.0
```

| Case Study              |           |                                                                  |                                                |                                         |                                                      |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|
| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  |
| Understanding of Case   | 25        | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |

|          |    |                                                      |                                      |                                   |                                      |
|----------|----|------------------------------------------------------|--------------------------------------|-----------------------------------|--------------------------------------|
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
|----------|----|------------------------------------------------------|--------------------------------------|-----------------------------------|--------------------------------------|

### Code of Conduct

*I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: KS  
THARUNESHWAR

### RUBRICS FOR CALCULATION

|                 |    |                                                          |                                               |                                         |                                       |
|-----------------|----|----------------------------------------------------------|-----------------------------------------------|-----------------------------------------|---------------------------------------|
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided     |
| Clarity         | 10 | Clear, well-structured, and easy to understand           | Organized and understandable presentation     | Limited clarity and structure           | Poor clarity; difficult to understand |

### Mark Evaluation:

| Case Study          | Understanding of case (25) | Applications of concepts (25) | Analysis (20) | Solution Design (20) | Clarity (10) | Total Marks (Each - 50) |
|---------------------|----------------------------|-------------------------------|---------------|----------------------|--------------|-------------------------|
| Q1                  |                            |                               |               |                      |              |                         |
| Q2                  |                            |                               |               |                      |              |                         |
| Overall Total (100) |                            |                               |               |                      |              |                         |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Name: _____       | Signature: _____   | Signature: _____   |
| Signature: _____  |                    |                    |
| Date: _____       | Date: _____        | Date: _____        |